

Progress brief 01

AI/ML for missile detection, warning and tracking — SBIR topic DAF26BX06-NV510
24 September 2026 · written for Mason · plain-language snapshot plus a self-audit

The one-paragraph version. We have built a careful, well-checked *foundation*: a simulator of satellites, targets, atmosphere and background, plus two classical trackers proven correct against outside references. We have not yet built the part that turns a target's position into what a satellite *actually sees*, and we have not written any AI. We decided the proposal is about a new AI/ML tracking contribution; the literature showed the idea we first had (learning which models a tracker runs) is already published three times, so the contribution must be sharper. Before any AI work, we need a fair, leak-proof test bench. The self-audit at the end found real shortcuts — most seriously, **none of this work is saved in version control yet**.

1. What we are trying to do

The Space Force wants to know whether AI/ML can help satellites that watch for missiles in infrared light (called OPIR) detect, track and warn about them better. Phase I is not a product; it is a **feasibility study**: a simulation, analysis and plan showing an idea is worth funding. It is judged on four things — **accuracy, latency, resilience, coverage** — and it must show the AI is **explainable** and says **how confident** it is.

Everything is unclassified. Real missile-warning satellite data is secret, so everything we show will come from simulation built on public information. The government expects this for Phase I.

2. Where we are

Phase	What it is	Status
D — Data	Find, download, organise and understand every useful public data source	Passed 23 Sep
S — Simulation core	Build the simulated world and the classical trackers	In progress — 6 of 12 tasks done
B — Benchmark (proposed)	A fair test bench that any algorithm, classical or AI, is scored on	Proposed, not yet in the plan
M — The AI/ML contribution	Design, build and test the new method	Direction under discussion
T — The trade	Where should processing happen: ground, satellite or both?	Not started
W, X — Writing, submission	CONOPS, feasibility study, cost, upload	Not started
G, E — Compliance, outreach	Registrations, contacting the program officers, Q&A	Not started — see audit

Size of what exists: 23 code modules, 303 automated tests (all passing), 43 recorded decisions, 33 recorded plan changes, about 960 MB of kept data and 7.7 GB set aside.

3. The foundation, in plain terms

Think of the simulator as a stage with actors. We have built most of the stage and the actors, and checked each piece against something outside ourselves — a published table, another team's software, or real measurements. That checking is the most valuable thing we have: when we say a number, we can show where it came from.

Piece	What it does, simply	How we know it is right	Confidence
Orbit prediction	Works out where each satellite is at any moment	Matches 533 official test cases to within a tenth of a millimetre	High
Constellation builder	Lays out a fleet of satellites in organised rings	Its orbits match the real SDA satellites we can observe	High
Coverage	Asks "how many satellites can see this spot, from good angles?"	Reproduces a published CSIS think-tank finding independently	High
Target flight paths	Where a hypersonic glide vehicle goes	Taken from Berkeley's published simulator; matches the paper's range to 0.09%	High, but only one path saved
Atmosphere and infrared absorption	How much of the heat signal survives the air	Matches the standard atmosphere table and a known physical fact about CO ₂	High
Earth background	What the target must be picked out against	Real GOES weather-satellite images	Real data, but the wrong colour of infrared (makes things look harder than they are)
Viewing geometry, track-quality scores	How good a two-satellite fix is; how we grade a tracker	Maths is standard, but not yet checked against an outside worked example	Medium
Classical tracker (IMM)	The standard way to follow a manoeuvring target	Agrees with an independent open-source version to 13 decimal places	Very high
Smarter classical tracker (LMS)	A version that only runs the models it needs	We re-ran a 2000 paper's own experiment and got within 1.5% of its published numbers	High (two known gaps — see audit)

What the foundation has already taught us

- **Speed of light is not the bottleneck.** Signals travel from satellite to ground in hundredths of a second; warnings take tens of seconds. So "process on the satellite to be faster" only wins if the processing itself is faster there — not because the satellite is closer.
- **Comparing consecutive images lets you see about 13× fainter things** than looking at one image, and faster revisits make targets visible that are otherwise hidden.
- **The "smarter" tracker is cheaper, not more accurate.** A survey claimed it was much better. The original paper actually shows it is slightly *less* accurate and about half the cost. A second, independent paper agrees. We now trust primary sources over summaries.

- **Almost every latency number anyone produces is assumption** (over 99.5%), so we should report comparisons and break-even conditions, not absolute times.

4. Data: what we have, and is it enough?

Data	Why we need it	What we have	Enough?
Satellite orbits	Where the watchers are	Current and historical public orbit data	Yes
Constellation designs	Which fleets to test	Published SDA and CSIS designs	Yes (notional)
Target flight paths	What we are tracking	Berkeley simulator — but one saved path	No — need a family of paths
How bright targets are	Can they be seen at all?	Published estimates — which disagree by 9–13×	As good as public gets; unresolved
Earth background images	What hides the target	A handful of real GOES frames	Enough to start; wrong band; few scenes
Atmosphere, Sun, Moon, Earth rotation	Physical corrections	Standard sources	Yes
Real tracking data (aircraft)	Real motion to test the tracker on	1 hour, 13,000 aircraft (OpenSky)	Yes — but never used yet
Sensor details (noise, detection odds, false alarms, frame rate)	What a satellite actually reports	Not public. Must be swept over declared ranges	Cannot be better; must be stated honestly
Real missile observations	Ground truth	Do not exist publicly	Impossible for anyone at Phase I

Verdict. We have enough to build a credible *simulation-based* feasibility study. We do *not* have, and cannot get, what would support claims about real operational performance — nobody outside the classified world can. The gaps that are **in our control**: a proper family of target flight paths, the sensor model, actually using the aircraft data, and detecting at least one real rocket launch in the GOES images.

5. What is "learned model-set adaptation"?

The tracker as a committee. A manoeuvring target might fly straight, turn left hard, climb, or turn right gently. A classical tracker (the IMM) runs a *committee* of forecasters, each assuming one kind of motion. Every time a new observation arrives, each forecaster is scored on how well it predicted it; the committee's answer is a weighted blend, and the weights are themselves useful: "80% sure it is turning left."

Fixed versus variable committee. The basic version consults the whole committee every time — reliable but expensive. The "variable-structure" version only invites the members who seem relevant right now, plus the neighbours of whoever is currently winning. LMS is a specific rulebook for that: *drop anyone below 0.01% belief; if anyone is above 30%, invite the manoeuvres next to theirs.*

"Learned" model-set adaptation means replacing that hand-written rulebook with a trained AI policy that decides who gets invited each moment.

Where that idea stands. Three research groups have already published it (2023, 2024, 2025). On honest tests it gives a few percent. None of them checked whether the tracker's *confidence* stays trustworthy afterwards, none used satellite-style angle-only sensing, and none tested on motion patterns the AI had never seen. Those gaps are real, but the idea itself is not new.

6. The end-to-end process, and where we would build

#	Stage	Plain meaning	Status
1	Truth	Where the target and satellites really are	Built
2	Sensing	What a satellite reports: directions (no distance), with noise, misses and false alarms	Missing — the main blocker
3	Detection	Picking the target out of the background	Physics built; not joined up
4	Tracking and fusion	Turning many looks from many satellites into one track	Trackers built and proven, but only on simple position data, not satellite angles
5	Confidence	How good is the track, and does it know how good it is?	Scoring defined; one key metric not yet wired in
6	Warning timeline	How long from launch to a decision	Model built; mostly assumption by nature
7	Architecture trade	Process on the ground, on satellites, or both?	Not started

Where the AI would go. Stage 4 is the natural home — tracking a target that manoeuvres, seen only as directions from moving satellites. Stage 3 (detection in images) is the alternative.

What must come first regardless. Stage 2 and a fair test bench ("Phase B"): many kinds of simulated flights, whole categories held back so the AI cannot memorise the simulator, the same random conditions for every method, error bars on every comparison, and the classical tracker tuned as hard as the AI. Without this, any AI result is unfalsifiable.

7. Our options

Option	In short	Upside	Risk
A. Sharpened incremental	A learned committee-selector that keeps the tracker's confidence trustworthy, tested properly on satellite-style data	Defensible, falsifiable, clearly answers the topic's confidence and explainability asks	Modest novelty; expected gain is a few percent
B. New algorithm	Describe manoeuvres the way a hypersonic vehicle actually flies (bank angle, lift) instead of a grid of accelerations; AI learns how the controls tend to change	Potentially much larger payoff; far better explanations ("banking 40°, ~2 g")	Novelty unverified; may not be observable from angles alone; slower
C. AI detection	AI finds faint targets in the infrared images	We hold real imagery	Wrong infrared band; a different pipeline

A is the safety net for B. Both need Phase B first. The next step proposed was a third targeted research brief to test whether B is genuinely new and genuinely observable.

8. Self-audit: where we cut corners

You asked whether we have taken serious shortcuts. Yes, several. In order of seriousness:

1. **No version control.** All 23 modules, 303 tests, the plan and the analysis exist only as files on disk — zero commits. One bad overwrite or disk problem loses everything, and we cannot show a reviewer the history. This is the single most serious gap and the cheapest to fix.
2. **Second-hand sources, three times.** I built the first VS-IMM from a survey's description instead of the paper; recorded a "failed replication" of a claim I had not traced; and called learned model-set adaptation an "open problem" without searching. Each was caught, but the pattern is mine and it is recurring.
3. **Research reports largely unverified.** The second deep-research report has about 90 citations, none of them resolvable as delivered; I checked 4. Its conclusions shaped our plan.
4. **"Done" that is thinner than its name.** Target motion is marked done with one flight path. The data phase passed in roughly a day across 24 domains; "understood" was claimed for aircraft data we have never run through code.
5. **The replication has two loose ends** we explained rather than investigated: our random scenario is 13% easier than the paper's, and the paper's accuracy gain for its "forgetting factor" variant only partly reproduced.
6. **Compliance and outreach untouched.** The window to contact the program officers directly closed on 23 September. Registrations that gate submission have not started.
7. **Housekeeping drift.** The project instructions still say we are in the data phase; the assumptions register has 26 rows for far more numbers than that; there is no one-command figure generator yet.

9. Where I would dig deeper, with targeted research

What to re-examine	Why it matters	How to settle it
The LMS replication's loose ends	It is our baseline; unexplained gaps weaken everything built on it	Acquire Li & Zhang's reference [29] (their Part IV paper, where this benchmark was designed, likely <i>IEEE T-AES</i> 1999) and the 1998 SPIE version; research prompt: <i>"exact specification of the Li-Bar-Shalom 13-model benchmark — sampling time, initialisation, random-scenario generator, forgetting-factor rule"</i>
The two research reports	They changed our decisions; most citations are unchecked	Get the bibliography export; run a verification pass that labels each load-bearing claim verified / not found / contradicted
Modules with no outside check	Viewing geometry and track-quality scores will grade every algorithm	Check against a published angles-only error bound and a textbook NEES worked example; research prompt for public worked examples with numbers
How bright a glide vehicle looks	Whether glide-phase detection is feasible at all rests on a 9–13× disagreement	Targeted prompt: <i>"all public estimates of HGV hardbody infrared intensity, their assumptions, and what drives the disagreement"</i> ; sweep both ends in the simulation

Real-data anchor	Converts "we simulated it" into "we did it on real imagery"	Detect one real, publicly scheduled rocket launch in GOES images (task D-05-d)
Gate D depth	A one-day pass over 24 domains	Re-walk the three domains that feed the AI work most (target paths, background, tracking data) and exercise each in code

10. Accountability checklist — this edition's answers

Standing questions live in `progress_briefs/accountability_checklist.md` and are answered at the end of every brief. Status: **OK** · **WATCH** (known, contained) · **ACT** (needs a plan task before next brief).

#	Question	Answer	Status
1	Load-bearing claims traced to primary sources we read?	Tracking claims: yes (Li & Zhang, three MSA papers read in full). Detection and IR-brightness claims: partly, via earlier transcription	WATCH
2	Claims resting on unverified research output?	Yes — report 02, 4 of ~90 citations checked; report 01 partly checked	ACT
3	"Nobody has done X" without searching?	Yes once (learned MSA). Caught and retracted	WATCH
4	Modules without an outside check?	Viewing geometry, track quality, limb geometry, GOES detection floor	ACT
5	Unexplained replication differences?	Random scenario 13% easier; forgetting-factor variant only partly reproduced	ACT
6	Method changed after seeing results?	Yes: one-step-back was added after the first run — but the paper states it was used, so justified by source. Iterated expansion tested and reported as no effect	WATCH
7	Assumptions all registered?	No — 26 register rows, many more numbers	ACT
8	Conclusions that flip with a parameter?	Glide-phase detectability flips across the 9–13× brightness range. No sensitivity analysis run yet	ACT
9	"Done" thinner than its name?	Target motion (1 path); data-phase "understood" for unused aircraft data	ACT
10	Acquired but unused data?	OpenSky aircraft data (495 MB); most GOES frames beyond the first test	WATCH
11	Skipped because hard or blocked?	Sensor model (S-07); association (JPDA); real-launch detection; Tap Lab library — all recorded	WATCH
12	Contribution and closest prior art?	Undecided between options A and B. Closest prior art for A: Lu et al. 2025, read in full. For B: not yet searched	WATCH
13	What would falsify the main claim, and tested?	For A: confidence (NEES) breaking down under the learned selector. Not tested — nothing built yet	WATCH
14	Version control and one-command rebuild?	No commits at all; no figure generator	ACT

15	Plan and instructions current?	CLAUDE.md says phase D; Phase B not yet written into the plan	ACT
16	Gate reviews real?	Gate D was reviewed with you, but at the level of my summaries rather than source by source	WATCH
17	Externally clocked items?	All untouched; direct program-officer contact window closed 23 Sep	ACT
18	Still aimed at the topic's metrics?	Yes; the confidence/explainability focus came straight from the topic text. Resilience and coverage are furthest from the AI work	OK
19	Where to dig?	Section 9	—

Suggested order before any algorithm work: (1) put the project under version control; (2) fix the plan and instructions drift; (3) verify the research reports' load-bearing claims; (4) close the replication loose ends; (5) then write Phase B into the plan and build the sensing stage. Your call on all of it — nothing above has been started.